EXPLORE INNOVATE EXCEL

Department of Metallurgical and Materials Engineering

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SYNOPSIS

Title: Machine Learning-Assisted Prediction and Interpretation of Adsorption Capacity of Carbon-Based Materials for Water Pollutant Removal

Aim:
To remove toxic heavy metals (eg.Pd,Cr) from polluted water using carbon-based adsorbents, and to predict/optimize adsorption capacity using machine learning (e.g. Random Forest, XGBoost) trained on lab and literature data — addressing CO1 and supporting clean-water goals.

Objectives:
- Identify heavy-metal contamination of water (e.g., Cr, Pb, Cd) as a real-world industrial and environmental engineering problem, and review its major industrial sources and impacts.
- Conduct a literature survey of carbon-based adsorbents and existing machine-learning approaches for adsorption-capacity prediction, to identify research gaps and align objectives with industrial, societal, and environmental needs.
- Prepare and characterize carbon-based adsorbents (activated carbon, biochar) using standard techniques.
- Run batch adsorption experiments and build a combined lab-generated + literature dataset.
- Train, evaluate, and interpret ML models (e.g., Random Forest, XGBoost, SHAP) to predict and optimize adsorption capacity.
- Validate model predictions experimentally and assess the approach's relevance to industrial-scale water treatment and societal/environmental clean-water needs.

Facilities required:
Adsorbent synthesis: muffle/tubular furnace for carbonization and activation, drying oven, ball mill/sieve shaker, precision analytical balance. Adsorption studies: pH meter, orbital shaker/magnetic stirrer, centrifuge, and Atomic Absorption Spectrophotometer (Optional) for accurate quantification of residual heavy-metal concentration.

Analysis tools required:
Python (Pandas, Scikit-learn, XGBoost, SHAP) for data analysis and ML modelling; Excel/CSV for data management; Git/GitHub for version control; Word/LaTeX for reporting.

Student Name	Signature	Remarks

Mentor signature:

Date: